Design and Simulation of an Adaptive Underactuated Gripper for Grasping Lunar Rock

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***Abstract*—Robotic grasping by grippers is a representative operation conducting scientific activities in space, such as on-orbit construction, on-orbit maintenance, and planetary exploration. The underactuated gripper is an emerging research direction of space robots. This paper proposes a new configuration of the adaptive underactuated gripper for grasping lunar rocks. In this paper, based on the contact conditions of different stages under ideal conditions, the kinematic analysis of the equivalent mechanism at each stage is carried out. The gripper adopts static equilibrium and dynamics calculation and can successfully secure a spherical object with a diameter of 100 mm to 200 mm. According to the analysis, the numerical simulation shows that the underactuated gripper has a good accessible space and a stable grasping process. This underactuated gripper is proved numerically to have the adaptative morphing capacity to tighten on the target with multiple advantages, such as few actuation units, low cost, good load capacity, and wide grasping range.**

Keywords— underactuated gripper, self-adaptive, kinematics, space mechanism, lunar exploration

I. INTRODUCTION

The moon is an important destination for planetary exploration missions. Its surface mainly comprises fragmented rocks, regolith, and dust [1-2]. Those geological materials contain much information of high scientific values, such as interplanetary meteorites, ejected chucks from the impact of an asteroid or comet, and signatures of space weathering and radiation. It may help us understand the evolutionary history of the solar system [3-5].

The sample acquisition of lunar rocks is the primary objective of scientific exploration. Relying on robots to complete the grasping of lunar rocks requires a robust design to mimic manual grasping. For space robotics underactuated grippers are an essential direction for the development of space robotics. The reliability of robotic systems is essential for space robotics [6]. The simplicity of structure and control is an important measure to increase the safety and reliability of the gripper. Overall, the gripper's size, weight, and power consumption should be designed as small as possible.

The underactuated gripper uses its structure, adaptive mechanism, and springs to access a broad grasping envelope to fasten and morph on the surface of geo-objects. This mechanism with fewer drive units is a simple structure providing a practical and efficient means to solve these challenges.

The underactuated humanoid TH-1 and TH-2 of Tsinghua University were developed by Zhang [7]. This gripper is designed with a spring-driven moving sub to complete the passive joints. Huang [8] designed an underactuated gripper, which is a linkage-driven prosthetic hand with an underactuated linkage mechanism for both the proximal and middle finger joints and a coupled linkage for the middle and distal finger joints to achieve a humanoid motion. These kinds of humanoid grippers have the concept of under-actuation but are incapable of adapting to objects due to their limited degrees of freedom (DOF). Most grippers take the configuration of three-finger or three-knuckle to achieve then under-actuation, rather than directly mimicking a human hand. Townsend designed the grasper of Barrethand [9]. The gripper is a three-finger structure with a DOF of seven and has a total of four drives. Three of the motors control each of the three fingers, and a motor controls the indexing motion of the fingers. Mei [10] developed two out-of-cabin service robot shape adaptive grippers. It is a three-finger gripper, and one or four motors are used to control the gripper to perform various grasping tasks in the cabin. The SARAH hand was developed by MDROBOTICS, Canada [11]. This gripper is a three-finger structure with a DOF of ten and requires only one drive for grasping through a mechanical limit and spring mechanism. The gripper serves the International Space Station and has a superior adaptability to targets with various shapes and sizes. There are also underactuated grippers with different transmission or drive configurations. A linkage mechanism couples the HIT/DLR gripper [12] to achieve the fingertip motion. It also has a series of bevel gear combinations to achieve the motion of the other two phalanxes. Cui [13] designed an underactuated gripper structure driven by two sets of four-link mechanisms. The gripper uses two sets of links, the driving working link and the parallel positioning link, to complete the grasping, which has good stability and an unchangeable grasping force. Due to the structure's complexity, the analysis method of underactuated grippers is also a problematic point of research. Dollar of Yale University [14-15] developed the SDM hand. The polymer was used to connect the gripper, and its joints are flexible, allowing for

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extensive reach and excellent grasping results. The hydrostatic model of an underactuated manipulator was established based on the principle of virtual work and introduced the zero-space basis vector of the manipulator Jacobi matrix to reduce the computational complexity. An underactuated finger mechanism was designed by Wu [16]. The gripper has a DOF of 3 and contains a torsion and a tension spring. They analyzed the kinematic equations of the adaptive finger mechanism on different targets using the law of least resistance.

In this paper, an underactuated gripper is designed for grasping lunar rocks, which can achieve passive adaption to the target, and has better operational performance than the common gripping device. Meanwhile, the underactuated gripper has better grasping capability than the fully driven gripper [17] because it requires fewer drive and control elements. The grasping motion of the underactuated gripper is related to both its mechanical structure and target contact: i) when the underactuated gripper does not contact the target, its motion is determined only by the mechanical structure and the driving force; ii) when the underactuated gripper contacts the target, its motion and the resulting reaction force are determined by all the factors of the interactive position, mechanical configuration, and driving force. The energized element in the underactuated gripper has a significant influence on the grasping process. However, there are fewer studies related to it, and the selection of spring parameters for many examples is obtained by taking advantage of numerical experiments but lacking systematic modeling. In general, the design of underactuated gripper configurations is complicated. The existing references are few and have drawbacks, such as large size and small grasping force. Therefore, a novel underactuated gripper needs to be developed.

The rest of this paper is organized as follows: Sec. 2 describes the structure and the grasping process of the proposed underactuated gripper in detail. Sec. 3 analyzes the kinematic motion of the single-finger mechanism in each stage. Sec. 4 applies the kinematic and dynamic simulation of the underactuated gripper. Finally, Sec. 5 concludes this study.

II. Proposed Underactuated Gripper

This paper proposes a three-finger underactuated gripper. The single-finger mechanism includes eight linkages with two tension springs. The DOF of each finger mechanism is

$$F = 3n - 2P_{\mathrm{L}} - P_{\mathrm{H}} = 3\times 9 - 2\times 12 - 0 = 3 \qquad (1)$$

As the three-finger mechanisms are identical, the total DOF of the whole gripper is 7. A single motor drive drives the screw to rotate and realize the linear motion, thus making each finger move until it begins to touch with the target and the envelope grasping of the target is completed.

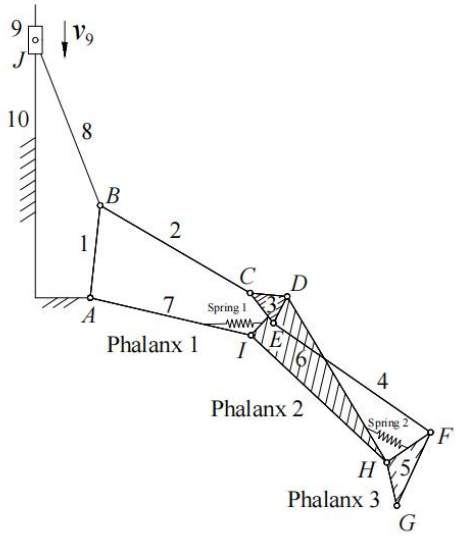


Fig. 1. Schematic diagram of the finger mechanism.

In Fig. 1, link 10 indicates the frame, slider 9 indicates the screw that is the only drive, links 1 to 8 form the finger mechanism, spring 1 is connected between links 6 and 7, spring 2 is connected between links 5 and 6, link 7 indicates the phalanx 1, link 6 indicates the phalanx 2, and link 5 indicates the phalanx 2, A to I are all rotating joints, J is the prismatic joint.

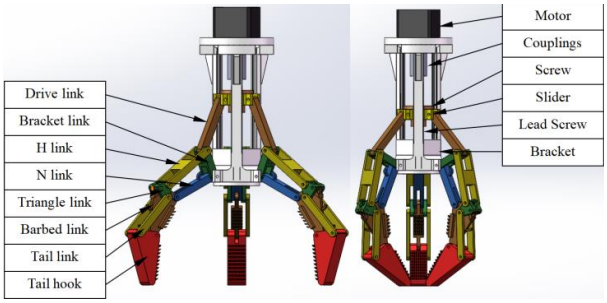


Fig. 2. Structure of the proposed underactuated gripper.

The process of grippers grasps the object are: 1) the motor rotates in the reverse direction, the screw moves upward, then all three-finger mechanisms are fully opened; 2) the system approaches the target object, and the bottom of the bracket approaches the object surface; 3) the motor rotates in the forward direction, the screw moves downward, the N link rotates around the axis of the bracket, and the three-finger mechanisms begin to close; 4) when the N link makes contact with the object surface, the barbed link continues to rotate around the axis of the N link; 5) when the barbed link and the object surface contact, the tail hook continues to rotate around the barbed link axis and further close; 6) finally the tail hook contacts the object surface and pierce into the object. The tail hook and the barbed link can be a bit into the hard surface of the target rock, increasing the contact friction and improving the gripper's grasping ability.

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III. KINEMATIC ANALYSIS

A. Kinematic definition

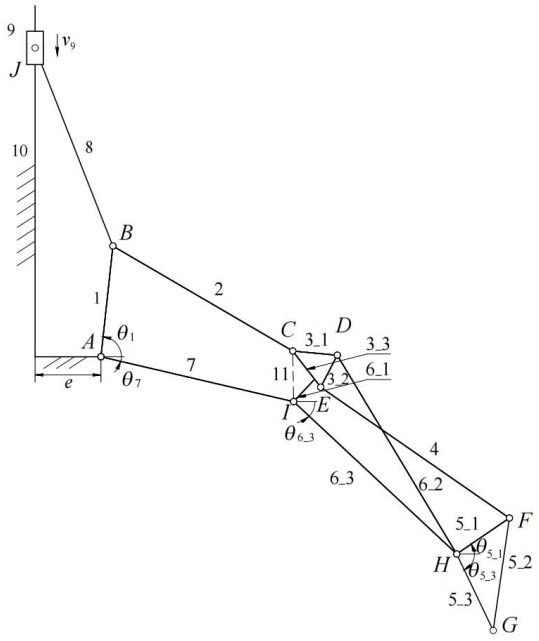


Fig. 3. Parameter definition of the finger mechanism.

The finger mechanism consists of 14 links and a slider. Due to the geometric relationship constraint between the components, the analysis in Fig. 3 shows that only 13 variables are needed to determine the shape parameters of the finger mechanism. In this paper, the bar length of l_1, l_2, l_{3_2}, l_4, l_{5_1}, l_{5_3}, l_{6_1}, l_{6_2}, l_{6_3}, l_7, l_8, the distance of e, and the angle of θ_{5_3} are selected as geometric parameters of subsequent design and analysis.

The initial opening position of the underactuated gripper not only affects the range of motion of individual phalanxes but also impacts the spring's design. Hence, the design of the initial position is crucial for the design of the whole mechanism. The analysis of Fig. 3 shows that only five variables are needed to determine the initial position parameters of the finger mechanism. In this paper, l_{11}, θ_1, θ_{5_1}, θ_{6_3}, and θ_7 are selected as the initial position parameters, where l_{11} indicates the distance between joint C and joint I. The underactuated gripper's shape and initial position parameters are designed in Table I and Table II, respectively.

TABLE I. PARAMETERS OF FORCE

Shape parameters	l_1	l_2	l_{3_2}	l_4	l_{5_1}	l_{5_3}
Value	35 mm	65 mm	11.3 mm	72 mm	20 mm	40 mm
l_{6_1}	l_{6_2}	l_{6_3}	l_7	l_8	e	θ_{5_3}
20 mm	72.8 mm	70 mm	62 mm	60 mm	17.98 mm	-60°

TABLE II. PARAMETERS OF FORCE

Initial position parameters	l_{11}	θ_1	θ_{5_1}	θ_{6_3}	θ_7
Value	16.9 mm	97°	47.53°	-30°	-10°

The three-finger mechanisms of the underactuated gripper are identical and evenly distributed along the circumference, with the center as the axis. Each finger mechanism moves in its plane, so it is sufficient to analyze any finger mechanism. The finger mechanism can be divided into two parts, the drive module and the phalanx module, for separate kinematic analysis.

The equivalent mechanism of the drive module is the crank slider mechanism; the motion of the phalanx module is related to the driving force and the position of the contact points, and the motion of each joint follows the law of least resistance.

B. Kinematic Analysis of Drive Module

The drive module consists of a slider, a filament, a slider, link 8, and link 1, which can be considered an eccentric crank-slider mechanism for kinematic analysis.

C. Kinematic Analysis of Phalanx Module

There are three assumptions for the finger mechanism to grasp the object: i) both the linkage mechanism and the target object are rigid and do not produce deformation during the grasping process; ii) the contact point between the finger mechanism and the grasped object is always constant during the envelope grasping process; iii) the stiffness of the two springs needs to satisfy: the stiffness is sufficiently large to keep the initial stretching state still until it touches the object, and spring 1 should meet the law of least resistance before spring 2.

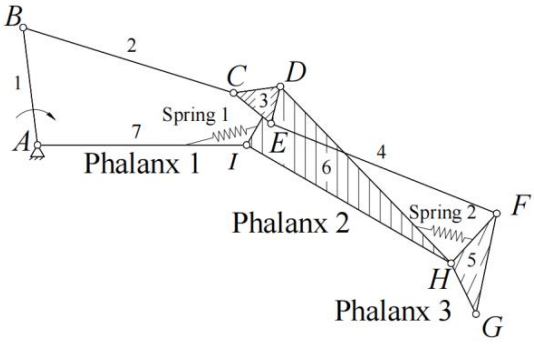


Fig. 4. Phalanx module.

Since the finger-rock interaction affects the finger mechanism's motion, the grasping process can be divided into three stages by the contacting points of three phalanxes and the target object. Therefore, it is easy to analyze the finger mechanism through its equivalent mechanism in each stage [16].

1) The first stage

The first stage is the motion of the finger mechanism before it touches the target object. According to the assumption iii, before touching the object, both spring 1 and spring 2 remain in the same initial stretched state because the stiffness of the two tension springs is large enough. Therefore, the mechanism can be equivalently considered a rigid body at this stage.

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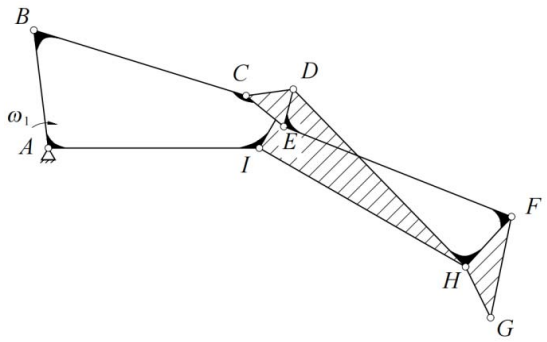


Fig. 5. Equivalent mechanism of the first stage.

2) *The second stage*

The second stage is when phalanx 1 (link 7) touches the target object, and phalanx 2 (link 6) does not touch the target object. According to the assumption ii, phalanx 1 (link 7) does not move after contacting the target object and can be seen as a rack; according to the assumption iii, the damping of deformation of spring 1 in this stage is less than the damping of other rotations. It means spring 1 deforms while spring 2 remains unchanged in its initial stretched state. Link 11 is the auxiliary line *CI*; the motion process of the second stage is equivalent to the four-linkage mechanism *ABCI* with link 1 as the active link, whose DOF is one.

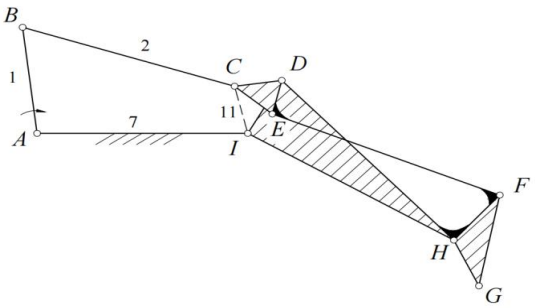


Fig. 6. Equivalent mechanism of the second stage.

3) The third stage

The third stage is when phalanx 2 (link 6) touches the target object, and phalanx 3 (link 5) does not touch the target object. According to the assumption ii, phalanx 1 (link 7) and phalanx 2 (link 6) do not move after contacting the target object and can be seen as a rack; according to the assumption iii, the damping of deformation of spring 2 in this stage is less than the damping of other rotations. Link 12 is the auxiliary line *AD*.

The motion process of the third stage can be equated to the combination of two four-linkage mechanisms *ABCD* and *DEFH*. Link 1 is the driving link of the four-linkage mechanism *ABCD*, and link 3 is marked as the driving link 3' of the four-linkage mechanism *DEFH*. Both of their DOF are 1.

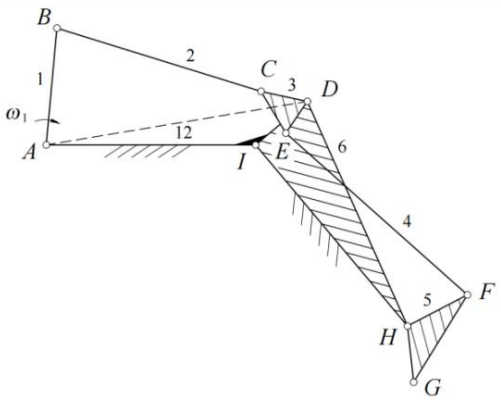
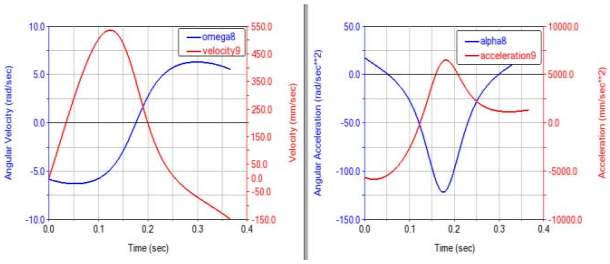


Fig. 7. Equivalent mechanism of the third stage.

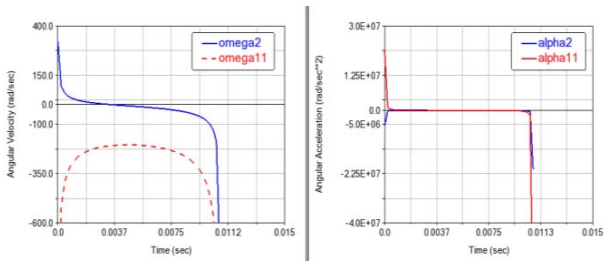
IV. Numerical Simulations

A. Kinematic Simulation

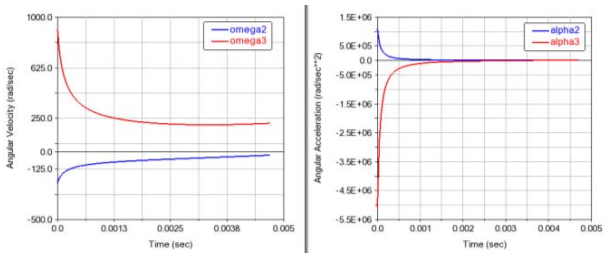
The selected parameters are substituted into the kinematic equations and simulated numerically. The angular velocity of the active pole is set as -100 rad/s and the angular acceleration is set as 0. Each stage's angular velocity and angular acceleration are obtained in Fig. 8, where the horizontal coordinate is the angle of the active pole's rotation. The legend omega *i* and velocity *i* refer to the angular and linear velocity of the *i*th component, and the legend alpha *i* and accelerate *i* refer to the angular and linear acceleration of the ith component. The motion curve is very smooth except for the large angular velocity and angular acceleration at the pole. The range of motion can be limited to the range of angular acceleration close to 0 by the mechanical limiting structure.



(a) Equivalent mechanism of drive module

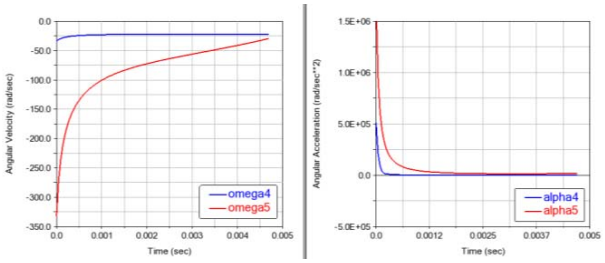


(b) Equivalent mechanism of the second stage



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(c) Equivalent mechanism *ABCD*



(d) Equivalent mechanism *DEFH*

Fig. 8. Motion curve of each equivalent mechanism.

The ranges of motion of phalanx 1, 2 and 3 are 167°, 210°, and 72°, respectively. Draw the accessible space of the gripper, and it can be concluded that it has a good range of motion.

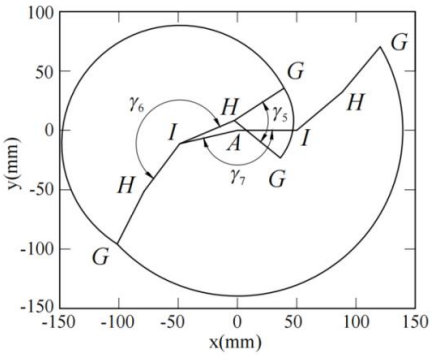


Fig. 9. Accessible space of the finger mechanism.

Considering that the gripper needs to be launched through the device when facing the moon rock grasping, the volume of occupied space should be calculated. When fully closed, the envelope of the grippe is a cylinder of 137 mm in diameter and 306 mm in height; when fully opened, the envelope of the gripper is a cylinder of 274 mm in diameter and 265 mm in height.

B. Dynamic Simulation

The loads on the model are the gravity, spring tension, the contact force between the gripper and the target object, and friction between links. In this paper, the effect of friction is not considered. Taking the gravitational acceleration of 1.63 m/s^2 on the moon, the contact force of each spring and phalanxes are selected in Table III.

TABLE III. PARAMETERS OF FORCES

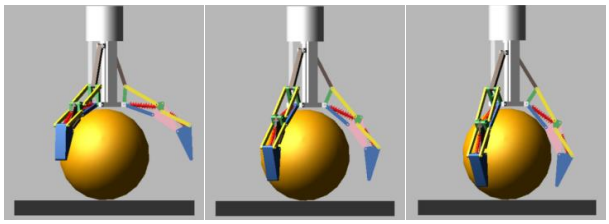
Contact Parameters	Each Phalanx	Spring Parameters	Spring 1	Spring 2
Stiffness(N/mm)	1×10^3	Stiffness(N/mm)	14	45
Force Exponent	2.2	Damping(Ns/mm)	2	2
Damping(Ns/mm)	10	Preload(N)	0.5	0.3
Penetration(mm)	0.1	Length at Preload(mm)	38.65	27.76

1) Static equilibrium calculation

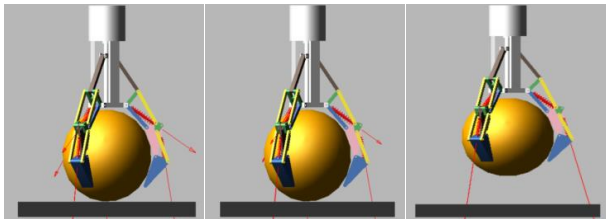
The underactuated gripper should remain in the opening state until it touches the object. Since the initial opening state influences the grasping process, it should be verified whether the finger mechanisms can stay still at the opening state when the slider keeps the initial position immobile. The simulation shows the system is in static equilibrium at t = 0 s.

2) Dynamics calculation

Using ADAMS for the dynamics simulation, the motion process of the gripper grasping the target object is obtained, as shown in Fig. 10.



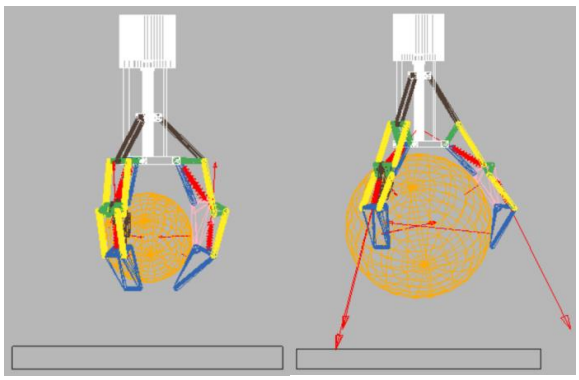
(a) t = 0 s (b) t = 2 s (c) t = 8 s



(a) t = 10.9 s (f) t = 12 s (g) t = 20 s

Fig. 10. Simulated grasping process of the gripper.

The gripper can grasp a particular range of different sizes of hypothetical spheres, as shown in Fig. 11. When it comes to the smaller objects, phalanx 3 is the first to contact the sphere, and the passive adapted joints driven by the driving force hold up the sphere; when it comes to the bigger objects, phalanx 1, 2 and 3 of the gripper contact the sphere in turn, and the sphere is mainly grasped by the contact force of the three phalanxes. As shown in Fig. 11, the gripper can successfully grasp the spheres with 100 mm to 200 mm diameters.

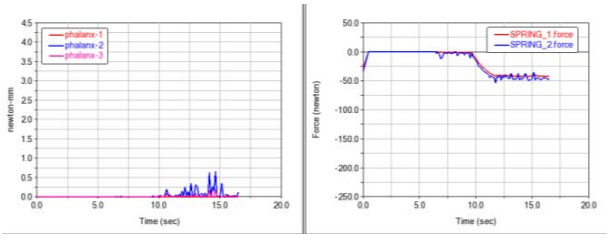


(a) d = 100 mm (b) d = 200 mm

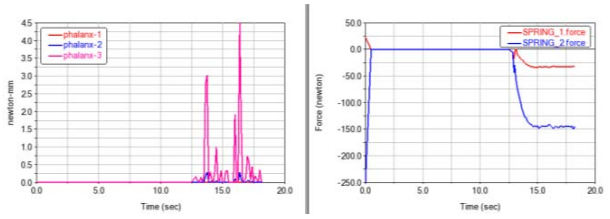
Fig. 11. Grasping cases of objects with the size of diameter of 100 mm and 200 mm.

The graphs of contact force versus spring force for each phalanx are shown in Fig. 12. From the graph, when the volume of the sphere is small, the contact force of phalanx 2 is much larger than the other two phalanxes; when the volume of the sphere is larger, the contact force of phalanx 3 is much larger than the other two phalanxes. This indicates that the underactuated gripper can achieve passive self-adaptation to the target shape in the grasping process. According to the results of a series of simulation experiments, the tension force on spring 1 is less than 100N, and the tension force on spring 2 is less than 200 N during the gripping process.

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(a) Contact force curve (d =100 mm) (b) Spring tension curve (d =100 mm)



(c) Contact force curve (d =200 mm) (d) Spring tension curve (d =200 mm)

Fig. 12. Force comparison between two objects with different sizes.

V. CONCLUSIONS

This paper proposes a novel adaptive underactuated gripper for automatically grasping moon rocks in lunar exploration missions. The DOF of the underactuated gripper is seven, and it contains three fingers with two tension springs, which can realize the passive grasping of the target under the drive of one motor. The proposed underactuated gripper is a compact structure and can minimize the consumption of valuable resources of volume, weight, and power in a space program. The underactuated gripper has fewer drive elements. Because of the unique feature of an underactuated mechanism, only one motor can control three or even more finger mechanisms to realize the adaptive grasping of different sizes of objects.

This paper proposes a method of kinematic analysis of the underactuated finger mechanism in the ideal state by assuming spring elements, simplifying the complex design process. The underactuated gripper can successfully grasp the hypothetical spheres with diameters of 100 mm to 200 mm in the rigid body dynamics simulation. The motion of the gripper components is smooth, and the contact force is stable during the grasping process. In summary, the underactuated gripper has the features of a small amount of driving elements, a simple structure, and a wide grasping range, which can be applied to sampling lunar rocks in deep space missions. Furthermore, the underactuated gripper may be evaluated for the grasping strength of brittle rocks and the adaptivity of different morphologies of rocks with corners and edges.

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